

MR-BSP Entropy Scope and Implementation Map

Issues 11–15: historical audit, simulation comparison, and definitive Visual-SLAM logic

ANPLOS / MR-BSP working note

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1 Conclusion

The Visual-SLAM implementation does not use a true joint-covariance entropy. Its planning score is a separable team score: the sum of the focused marginal entropy scores around the final positions of Robot A and Robot B. Therefore, if a realized-entropy column is intended to be the execution-time counterpart of a particular robot’s Step-1 prediction, it must again contain two terms, both calculated from that robot’s execution posterior:

$$H_{\text{real}}^A = h_A(b^{A,\text{exec}}) + h_B(b^{A,\text{exec}}), \quad H_{\text{real}}^B = h_A(b^{B,\text{exec}}) + h_B(b^{B,\text{exec}}).$$

The current rollout does not do this. Its “Robot A realized entropy” is only Robot A’s final-pose marginal, and its “Robot B realized entropy” is only Robot B’s final-pose marginal. The dashboard then adds those two values, even though they came from two different local beliefs. That arithmetic sum can be reported as a decentralized self-uncertainty performance statistic, but it is not the realized value of either robot’s Step-1 objective.

2 What the publication defines

The publication defines one cooperative objective, evaluated from the local belief of the robot performing the calculation:

$$J(b_k^r, a_{k+}).$$

The superscript r identifies the *belief perspective*; it does not turn J into a cost concerning only robot r . The state, observations, and actions are joint, and all robots use the same cooperative reward function. The Visual-SLAM terminal reward is written as

$$\rho_L(b_{k+L}^r) = -\beta H(b_{k+L}^r) - (1 - \beta) \mathbb{E} \left[d(g^r, \xi_{k+L-1}^r) + d(g^{r'}, \xi_{k+L-1}^{r'}) \right].$$

Thus, the distance component explicitly includes both robots. The entropy component is described only as $H(b^r)$, “the entropy of the belief.” The current manuscript does *not* explicitly state that this is implemented as the sum of two final-position marginal entropy scores. That specialization must be stated in the new resubmission if it remains the experimental metric.

The manuscript also assumes that the actions of other robots are available to robot r , while their observations may be only partially shared. This distinction determines which factors may legitimately appear in a robot’s execution posterior.

3 The implemented focused entropy

For a final pose X_i with 6×6 pose covariance Σ_i , the code keeps the translational 3×3 block and calculates

$$h_i = \frac{1}{2} \log \det(\Sigma_i^{\text{pos}}).$$

The Gaussian additive constant is omitted. Rotation uncertainty is also omitted. This is therefore an entropy-like focused uncertainty score, with the same action ordering as the corresponding Gaussian positional entropy when the dimensionality is fixed.

For each candidate joint action, the Step-1 planner selects the last pose of each robot and sums the two scores:

$$H_{\text{pred}}^r(a) = h_A(b^{r,\text{pred}})(a) + h_B(b^{r,\text{pred}})(a).$$

Both covariances in this equation come from the same predicted posterior belief b_{pred}^r maintained by the planning robot.

This is *not* the entropy of a full joint covariance. A true joint entropy would require the block covariance

$$\Sigma_{AB} = \begin{bmatrix} \Sigma_A & \Sigma_{AB}^{\text{cross}} \\ (\Sigma_{AB}^{\text{cross}})^\top & \Sigma_B \end{bmatrix}$$

and would depend on the cross-covariance between the endpoint poses. The implemented score deliberately ignores that cross-covariance and adds the two marginal scores instead.

4 What the current rollout calculates

For each robot r , the current execution evaluator:

1. loads the saved candidate posterior graph for the action selected by robot r ;
2. samples or loads a physical ground-truth realization;
3. generates image measurements by projecting known landmarks through the realized trajectory;
4. replaces candidate-path vision factors for the robot being evaluated;
5. optimizes the resulting factor graph;
6. extracts only that robot’s final-pose marginal covariance; and
7. converts only that covariance to focused entropy.

Ground truth is not itself an uncertainty distribution. It supplies the physical poses and landmarks used to generate measurements. The uncertainty comes from the posterior covariance after estimation:

$$\begin{aligned} \text{ground truth} &\longrightarrow \text{simulated measurements} \longrightarrow \text{posterior factor graph} \\ &\longrightarrow \text{marginal covariance} \longrightarrow \text{focused entropy.} \end{aligned}$$

The two current dashboard columns are therefore

$$E_A^{\text{current}} = h_A(b^{A,\text{exec}}), \quad E_B^{\text{current}} = h_B(b^{B,\text{exec}}).$$

The displayed sum is

$$E_{\text{self-sum}}^{\text{current}} = h_A(b^{A,\text{exec}}) + h_B(b^{B,\text{exec}}).$$

It combines two self-marginals obtained from two different local beliefs. It is a meaningful external team statistic if labelled that way, but it does not equal H_{real}^A or H_{real}^B .

5 Issue 12: historical Table 3 and execution-code audit

The historical implementation gives an unambiguous answer about the scope of the Visual-SLAM objective. Table 3 did *not* report a realized post-execution entropy. Its “Objective Values (J)” column reported the Step-1 planning-time hybrid objective of the chosen candidate joint action, computed from the reference robot’s belief. The reference robot was Robot A in the table-generation path.

For a candidate joint action a , the historical code calculated

$$H_{\text{pred}}^A(a) = h_A(b_{\text{pred}}^A(a)) + h_B(b_{\text{pred}}^A(a)), \tag{1}$$

$$\hat{J}(b_k^A, a) = -0.95 H_{\text{pred}}^A(a) - 0.05(\ell_A(a) + \ell_B(a)). \tag{2}$$

Here, both endpoint covariances are extracted from the *same* predicted posterior constructed from Robot A’s available belief. Each endpoint score is

$$h_i(b_{\text{pred}}^A(a)) = \frac{1}{2} \log \det \Sigma_{i,\text{final},xyz}^{A,\text{pred}}(a), \quad i \in \{A, B\}. \quad (3)$$

The path-length mode used for these results was `last`, so $\ell_A + \ell_B$ is the sum of the final motion segment for each robot. The stored `hybrid_obj` is the maximum of (2) across candidate joint actions. The table-generation code then selected the appropriate communication iteration for Baseline II, R-EnforceAC, or Baseline I and reported the mean and standard deviation across runs.

This historical metric is therefore best described as:

the sum of Robot A’s marginal focused uncertainty about Robot A’s final predicted pose and Robot A’s marginal focused uncertainty about Robot B’s final predicted pose, both conditioned on Robot A’s current information.

It is not a one-robot entropy, not a true joint entropy, not the sum of two independently computed self-entropies, and not an oracle calculation. Only Baseline I has full shared data; Baseline II and R-EnforceAC use Robot A’s belief at their respective communication states.

5.1 Numerical reproduction of the Table 3 value 11.35

For seed 0 at the Baseline-II communication state, historical saved data for the selected candidate contain

$$h_A = -6.0180656, \quad h_B = -6.1756823, \quad \ell_A + \ell_B = 4.7059834.$$

Consequently,

$$\begin{aligned} H_{\text{pred}}^A &= -6.0180656 - 6.1756823 = -12.1937479, \\ \hat{J} &= -0.95(-12.1937479) - 0.05(4.7059834) \\ &= 11.3487613 \approx 11.35, \end{aligned}$$

which exactly reproduces the Baseline-II entry in Table 3. At full communication, the corresponding saved value is 13.4300764, reproducing the reported Baseline-I value 13.43.

5.2 Why Table 3 is not evidence of realized entropy

The results pipeline copied `results[Robot_A][step_1][hybrid_obj]` into the per-iteration summary. Step-2 and Step-3 sampled objectives were used to test whether the Step-1 action remained preferred, but they did not replace the reported J -value. Table 3 therefore measures a predicted planning score for the selected action, not uncertainty recomputed after execution.

The historical file also contains a separate function named `get_entropies_post_exec`. Its intended scope is nevertheless clear: it constructs one post-execution belief from Robot A’s available data and one from Robot B’s available data, requests both final-pose covariances in each belief, and intends to return

$$H_{\text{post}}^A = h_A(b_{\text{post}}^A) + h_B(b_{\text{post}}^A), \quad H_{\text{post}}^B = h_A(b_{\text{post}}^B) + h_B(b_{\text{post}}^B).$$

Thus, the historical design agrees with the two-endpoint, one-belief-at-a-time scope adopted for the Issue-11-fixed experiments. It does not add Robot A’s self-entropy from b^A to Robot B’s self-entropy from b^B .

There is an implementation caveat: in the audited historical revision, `get_entropies_post_exec` called `cov_mat_to_focused_entropy` with one argument although the helper required the pose symbol and covariance matrix. This legacy post-execution path was separate from Table 3 and was not used to generate its J column. The defect affects whether that legacy function could execute successfully; it does not make its intended two-endpoint scope ambiguous.

6 Exact factor scope: legacy versus corrected execution logic

Consider Robot A evaluating execution after a planning history ending at $A20, B20$. Suppose Robot B communicated visual observations only through $B12$, while its observations from $B13$ through $B20$ remained private. Write $\mathcal{F}_{i,[u:v]}$ for all motion and visual factors of robot i over an interval, $\mathcal{M}_i^{\text{exec}}$ for execution motion factors, and $\mathcal{V}_i^{\text{exec}}$ for execution vision factors.

6.1 What is always valid in Robot A’s local execution posterior

Robot A may condition on:

1. its complete local history $\mathcal{F}_{A,[0:20]}$, including factors involving landmarks that were never shared with Robot B;
2. the communicated part $\mathcal{F}_{B,[0:12]}$ of Robot B’s history;
3. both robots’ actually executed motion factors $\mathcal{M}_A^{\text{exec}}$ and $\mathcal{M}_B^{\text{exec}}$, because the experiment assumes that the executed actions are known; and
4. all new camera measurements acquired locally by Robot A, $\mathcal{V}_A^{\text{exec}}$.

Robot B’s private historical vision factors from $B13$ through $B20$ must remain absent. Robot B’s execution vision factors may enter Robot A’s belief only if those measurements were explicitly communicated.

Accordingly, the corrected Robot-A factor set is

$$\mathcal{F}_{\text{exec}}^{A,\text{correct}} = \mathcal{F}_{A,[0:20]} \cup \mathcal{F}_{B,[0:12]}^{\text{shared}} \cup \mathcal{M}_A^{\text{exec}} \cup \mathcal{M}_B^{\text{exec}} \cup \mathcal{V}_A^{\text{exec}}. \quad (4)$$

Unless communication occurs after execution,

$$\mathcal{F}_{B,[13:20]}^{\text{vision}} \not\subset \mathcal{F}_{\text{exec}}^{A,\text{correct}}, \quad \mathcal{V}_B^{\text{exec}} \not\subset \mathcal{F}_{\text{exec}}^{A,\text{correct}}.$$

After optimizing this one Robot-A-local graph, the corrected score is

$$H_{\text{real}}^A = h_A(b_{\text{exec}}^A) + h_B(b_{\text{exec}}^A).$$

The Robot-B calculation is symmetric.

Ground truth or a simulated physical trajectory may be used offline to generate a noisy measurement. That operation does not automatically make the measurement available to every robot. A projection from Robot B’s realized pose becomes a factor in Robot A’s posterior only when the modeled communication protocol delivers that observation to A.

6.2 What the legacy post-execution function intended

The legacy `get_entropies_post_exec` construction appended the complete execution interval of both robots to each local post-execution belief. For Robot A, its intended factor set was effectively

$$\begin{aligned} \mathcal{F}_{\text{exec}}^{A,\text{legacy}} = & \mathcal{F}_{A,[0:20]} \cup \mathcal{F}_{B,[0:12]}^{\text{shared}} \cup \mathcal{M}_A^{\text{exec}} \cup \mathcal{V}_A^{\text{exec}} \\ & \cup \mathcal{M}_B^{\text{exec}} \cup \mathcal{V}_B^{\text{exec}}. \end{aligned} \quad (5)$$

Thus, the description *all of A’s data, B’s shared prefix, and both robots’ complete execution segments* correctly describes the legacy function’s intended graph. The gap $B13$ – $B20$ remained absent, but B’s execution vision factors reappeared after $B20$. This implicitly assumes that B’s new execution observations become available to A even though its older private observations do not.

That assumption is not valid for the corrected robot-local metric unless post-execution communication is explicitly part of the protocol. Moreover, the legacy function had the call-signature defect described above and did not produce Table 3.

7 Which logic corresponds to which experiment

Artifact or experiment	Quantity and belief scope	Factor and entropy logic
Published Visual-SLAM Table 3	Step-1 projected hybrid objective $\hat{J}$ from Robot A's belief	Sums the focused marginal entropies of A's and B's final <i>predicted</i> poses inside Robot A's candidate posterior, then combines that sum with both final path segments using weights 0.95 and 0.05. It is not realized entropy and does not use the legacy post-execution function.
Legacy post-execution path	Intended post-execution entropy, one two-endpoint result per robot-local belief	The <code>verify_against_bsln</code> and <code>get_entropies_post_exec</code> path uses the legacy factor scope in (5), including the peer's complete execution vision interval. The path was unused for Table 3 and contained a call-signature defect.
Moshe/Tanmoy search-and-rescue simulations	Whole binary-grid entropy inside each agent's local grid belief	Robot locations are known exactly. Planning marginalizes over transition, cell-state, and observation outcomes; execution samples a transition and a noisy cell observation. Bayesian cell updates replace pose-graph inference: there are no pose covariances, odometry factors, projection factors, or landmark variables.
Dashboard Experiment 22	Original 30-seed sequential-evidence operating point	Original realized columns contain only each robot's own final-pose marginal. Their displayed sum combines self-marginals from two different beliefs; it is not either robot's two-endpoint realized objective.
Dashboard Experiment 23	Original early instantaneous operating point	Uses the same original one-endpoint-per-perspective realized-entropy scope as Experiment 22.
Dashboard Experiment 24	Original late instantaneous operating point	Uses the same original one-endpoint-per-perspective realized-entropy scope as Experiment 22.
Dashboard Experiment 25: Definitive metric, execution-only landmarks included ($\chi = 1$)	Primary corrected variant of Experiment 22	Uses (4) separately for each belief perspective; adds both final-pose marginal focused entropies inside that one posterior; adds only the evaluating robot's execution vision; and admits landmarks that were invisible throughout both prior trajectories and are first triangulated from that robot's local execution images.
Dashboard Experiment 26: Definitive metric, execution-only landmarks excluded ($\chi = 0$)	Known-landmarks-only ablation of Experiment 25	Uses the identical Experiment-22 geometry, seeds, decisions, acceptance rows, and two-endpoint robot-local calculation as Experiment 25, but a landmark absent from the complete prior cannot enter the execution graph.
Dashboard Experiment 27: Experiment 24 (Issue 11 fixed)	Earlier corrected copy retained for provenance	Uses the corrected two-endpoint robot-local scope on the late instantaneous operating point; it predates the explicit $\chi = 1$ versus $\chi = 0$ pair.

Experiments 25 and 26 preserve Experiment 22’s planning decisions and acceptance mechanism exactly. Their only mutual difference is χ . Experiment 27 remains the earlier late-operating-point corrected copy. None of these turns realized entropy into the full hybrid J reported by the historical Table 3: no path-length term is added to the realized entropy columns.

8 Step-1 projected entropy versus realized entropy

They are not, in general, the same quantity. Two separate distinctions are important: metric scope and prediction versus realization.

8.1 Original Experiments 22–24: different scope

In the original experiment implementation, robot r ’s Step-1 projected entropy is

$$H_{\text{pred}}^r(a_r^*) = h_A(b^{r,\text{pred}})(a_r^*) + h_B(b^{r,\text{pred}})(a_r^*),$$

whereas its reported realized entropy is only

$$E_{r,\text{old}}^{\text{real}} = h_r(b^{r,\text{exec}}).$$

The old values therefore cannot be compared directly: the Step-1 value has two endpoint terms and the realized value has one.

8.2 Issue-11-fixed experiments: matching scope, different stage

The fixed realized metric uses the same two-endpoint scope:

$$H_{\text{real}}^r = h_A(b^{r,\text{exec}}) + h_B(b^{r,\text{exec}}).$$

Even after this correction, H_{pred}^r and H_{real}^r are not expected to be numerically identical.

Step 1 constructs a deterministic predicted posterior from the robot’s pre-execution belief, nominal candidate motion, and maximum-likelihood geometric observations generated from predicted poses and current landmark estimates. It produces one projected entropy value for each candidate action.

The realized calculation instead uses the action components actually executed, a sampled physical trajectory and landmark realization, noisy camera measurements generated from that realization, and only the information available to the evaluating robot. Repeating this process gives a Monte Carlo distribution summarized by a mean and standard deviation.

The comparison is therefore

$$\underbrace{H_{\text{pred}}^r(a_r^*)}_{\substack{\text{predicted before execution} \\ \text{nominal/MLE future observations}}} \quad \text{versus} \quad \underbrace{\mathbb{E}[H_{\text{real}}^r]}_{\substack{\text{estimated after simulated execution} \\ \text{actual action pair and noisy observations}}}.$$

At an action-inconsistent row, the difference is even more fundamental. The physical action pair contains A’s component of A’s decision and B’s component of B’s decision, so it need not equal either robot’s complete Step-1 proposed joint action.

The two values would coincide only under restrictive conditions: both robots execute exactly the joint candidate assumed in Step 1; the realized visible factor set equals the predicted factor set; the realized measurements equal their nominal maximum-likelihood values; both calculations condition on the same information; and nonlinear optimization produces the same linearization. In a linear-Gaussian model with a fixed factor structure, measurement values do not change the covariance, making equality more plausible. Visual SLAM is nonlinear and its visibility and linearization can change, so equality is not guaranteed even at an action-consistent row.

Consequently, the scientifically meaningful quantity is the prediction error or calibration gap

$$\Delta H^r = \mathbb{E}[H_{\text{real}}^r] - H_{\text{pred}}^r(a_r^*),$$

reported only after both terms use the same two-endpoint scope. A zero value means the predicted uncertainty was calibrated to the realized mean; it is not an identity imposed by the algorithm.

9 How to construct a valid execution posterior

For each robot r , construct b_{exec}^r from the information that is actually in that robot’s history. The construction should start from the robot’s real local belief immediately before execution, not from a Step-1 candidate posterior containing hypothetical future observations.

For Robot A, the valid construction is:

1. Preserve Robot A’s complete local motion and visual history.
2. Preserve only the portion of Robot B’s past visual observations that was actually communicated to A.
3. Preserve or append Robot B’s motion chain where its actions are known, in accordance with the paper’s action-availability assumption.
4. Append the motion factors corresponding to the actions actually executed by A and B.
5. Add every new visual observation genuinely acquired by A during execution, including A-local landmark observations that were never shared with B.
6. Add new observations acquired by B only if those observations were actually communicated to A by the evaluation time.
7. Optimize once, extract the marginal positional covariances of both final poses, and sum their focused entropy scores.

Robot B’s execution posterior is constructed symmetrically.

In the fixed implementation, saved pose values from the selected candidate may be used as an *initial guess* for nonlinear optimization of the newly appended trajectory. This is not an information factor: no candidate visual factor is copied, and the posterior information matrix is still defined only by the valid local prior, the executed motion factors, the evaluating robot’s available execution observations, and the existing weak numerical landmark priors. This distinction matters because pure odometric initialization can leave a large-residual visual-SLAM problem outside the convergence basin even when the factor graph itself is valid.

For numerical integrity, a measurement realization is never silently replaced merely because the default GTSAM marginalization rejects an ill-conditioned linear system. The evaluator first retries the *same* realization from the alternative candidate-based initial guess. If both nonlinear attempts remain ill-conditioned, it computes the local Laplace covariance from that same graph and linearization using a symmetric Moore–Penrose pseudoinverse of $J^T J$. The dashboard records how many realizations used each fallback. A new noisy realization is drawn only if all three calculations fail. The final fixed datasets require zero such rejected realizations.

9.1 The missing B12–B20 observations

Suppose A knows B’s motion history through B20 but has received B’s camera observations only through B12. Then B13–B20 pose variables and motion factors may legitimately exist in A’s belief, while B13–B20 private vision factors do not. The missing observations must remain absent. Their absence correctly causes A’s uncertainty about B to grow through the motion chain.

It would be invalid to generate B13–B20 image factors from ground truth merely to fill the gap. It would be equally invalid to begin inserting B’s private execution image factors after B20 unless those measurements were communicated. The fact that ground truth makes such projections possible for an offline evaluator does not make those factors part of Robot A’s information history.

10 What happens when the robots choose different actions

When Step 1 is inconsistent, Robot A’s chosen action index and Robot B’s chosen action index may refer to different proposed joint actions. The physical execution is not either complete proposal. Each robot executes its own component, producing a mixed realized joint action:

$$a_{\text{exec}} = \left(a_A^{\text{chosen by A}}, a_B^{\text{chosen by B}} \right).$$

Because the paper assumes that selected actions are communicated after planning and in parallel with execution, a post-execution belief may condition on this actual action pair. If the metric were instead evaluated before the peer action became known, the evaluator would have to marginalize over the unknown peer action rather than silently substitute either robot’s prediction.

11 Why the current saved posterior cannot simply be reused

The Step-1 candidate graph contains predicted motion and maximum-likelihood geometric factors for both robots’ candidate paths. The current rollout removes and regenerates candidate vision factors only for the pose prefix being evaluated. Hypothetical peer future factors can therefore remain in the graph and can indirectly change the evaluated robot’s covariance through shared landmarks.

Consequently, the correct two-endpoint calculation is not merely:

“Ask the current rollout graph for the other final covariance and add it.”

The evaluator must begin from the pre-execution local graph, strip all hypothetical future measurements, add the actual motion pair, and add only the observations available to the relevant robot.

12 Three distinct metrics that must not be conflated

Metric	Definition	Interpretation
Robot-A-perspective realized team score	$h_A(b^{A,\text{exec}}) + h_B(b^{A,\text{exec}})$	Realized counterpart of Robot A’s Step-1 predicted entropy.
Robot-B-perspective realized team score	$h_A(b^{B,\text{exec}}) + h_B(b^{B,\text{exec}})$	Realized counterpart of Robot B’s Step-1 predicted entropy.
Decentralized self-entropy sum	$h_A(b^{A,\text{exec}}) + h_B(b^{B,\text{exec}})$	External system statistic formed from each robot’s uncertainty about itself; not one robot’s J .
Centralized full-data score	$h_A(b^{\text{central}}) + h_B(b^{\text{central}})$	Oracle or Baseline-I-style benchmark using all observations from both robots; not an R-VerifyAC robot-local posterior.
True joint entropy	$\frac{1}{2} \log \det \Sigma_{AB}$, up to the Gaussian constant	Uses the full endpoint block covariance including cross-correlation; not computed by the current planner.

13 Definitive realized-entropy convention agreed with Moshe

The final decision is to evaluate execution strictly from the information available to the robot whose result is being reported. There is no oracle posterior and no cross-belief sum. Let s be the last communicated pose index and let $+$ denote the end of execution (for example, A_{30} and B_{30}). Write

- $\mathcal{M}_i^{0:+}$ for all motion factors of robot i from its initial pose through the end of execution;
- $\mathcal{V}_i^{0:+}$ for all actual vision factors acquired by robot i over the same interval; and
- $\mathcal{V}_i^{0:s,\text{shared}}$ for the prefix of robot i 's vision factors that was communicated to the peer.

Under the paper's assumption that executed actions are known, both robots know both complete motion chains. Under partial observation sharing, each robot knows its own complete vision history but only the communicated prefix of the peer's vision history. Therefore Robot A's execution graph is

$$\boxed{\mathcal{F}_A^{\text{exec}}(\chi) = \mathcal{M}_A^{0:+} \cup \mathcal{V}_A^{0:++}(\chi) \cup \mathcal{M}_B^{0:++} \cup \mathcal{V}_B^{0:s,\text{shared}}} \quad (6)$$

and Robot B's mirror computation is

$$\boxed{\mathcal{F}_B^{\text{exec}}(\chi) = \mathcal{M}_B^{0:++} \cup \mathcal{V}_B^{0:++}(\chi) \cup \mathcal{M}_A^{0:++} \cup \mathcal{V}_A^{0:s,\text{shared}}}. \quad (7)$$

In particular,

$$\mathcal{V}_B^{s+1:++} \not\subset \mathcal{F}_A^{\text{exec}}, \quad \mathcal{V}_A^{s+1:++} \not\subset \mathcal{F}_B^{\text{exec}}.$$

Shared-landmark variables and their communicated projection factors may connect the two motion chains inside either local graph. What matters is factor provenance: the factor must have been measured locally or received through the communication protocol.

13.1 Optional inclusion of landmarks first discovered during execution

The switch $\chi \in \{0, 1\}$ controls whether a robot includes landmarks that were absent from every pre-execution belief but were first detected by its own camera during execution:

$$\mathcal{V}_i^{0:++}(\chi) = \begin{cases} \mathcal{V}_{i,\text{known}}^{0:++}, & \chi = 0, \\ \mathcal{V}_{i,\text{known}}^{0:++} \cup \mathcal{V}_{i,\text{new}}^{\text{exec}}, & \chi = 1. \end{cases}$$

The primary real-execution interpretation is $\chi = 1$, because these are real measurements acquired by the evaluating robot. The $\chi = 0$ setting is a controlled known-landmarks-only ablation. The switch is applied symmetrically: A may add newly detected A-side landmarks only to b_A^{exec} , and B may add newly detected B-side landmarks only to b_B^{exec} , unless the corresponding measurements are later communicated.

13.2 Why the missing peer vision factors cannot be manufactured

Robot A cannot construct a realized observation from B_{18} or B_{22} merely because its estimate predicts that a landmark should be visible:

1. Projecting Robot A's maximum-likelihood estimates of the peer pose and landmark would create a *predicted pseudo-measurement*. That is a legitimate Step-1 planning operation, not an execution measurement.
 2. Projecting the hidden simulated ground-truth peer pose and landmark would create a physically plausible Robot-B measurement, but making it available to Robot A would leak oracle information across the stated communication boundary.
 3. In a real experiment, the measurement could originate only from Robot B's sensor. By assumption, Robot A has no access to that sensor stream unless B communicates the observation.
- Mixing ground-truth-generated local factors with estimate-generated peer pseudo-factors and calling both "realized observations" would therefore mix planning and execution semantics.

Ground truth is used only by the simulator's generative side:

$$(X_i^{\text{GT}}, L_j^{\text{GT}}) \xrightarrow{\text{pinhole projection} + \text{sensor noise}} z_{ij}.$$

The estimator receives z_{ij} and the noise model, never X_i^{GT} or L_j^{GT} . Consequently, only the robot that acquired z_{ij} may insert its projection factor before communication.

13.3 One optimization and two focused marginals per robot

Each robot optimizes its own complete execution graph exactly once:

$$\hat{\Theta}_r = \arg \min_{\Theta} \sum_{f \in \mathcal{F}_r^{\text{exec}}(\chi)} \|e_f(\Theta)\|_{\Omega_f}^2, \quad r \in \{A, B\}.$$

From the optimized local posterior, it extracts the translational marginal covariances of *both* final robot poses:

$$\Sigma_{A+}^r = \text{MargCov}_{b_r^{\text{exec}}}(A_+)_{xyz}, \quad \Sigma_{B+}^r = \text{MargCov}_{b_r^{\text{exec}}}(B_+)_{xyz}.$$

The two focused entropy terms and their sum are

$$\begin{aligned} h_A^r &= \frac{1}{2} \log \det \Sigma_{A+}^r, & h_B^r &= \frac{1}{2} \log \det \Sigma_{B+}^r, \\ H_{\text{real}}^r &= h_A^r + h_B^r, & r &\in \{A, B\}. \end{aligned} \tag{8}$$

Thus H_{real}^A and H_{real}^B will generally differ. Typically A knows its own final pose better than B knows A’s final pose, and B knows its own final pose better than A knows B’s final pose. This difference is expected evidence of decentralized information, not an inconsistency in the metric.

For naming precision, H_{real}^r is the raw realized entropy score. The corresponding entropy reward is $-H_{\text{real}}^r$. If the paper’s full hybrid objective is reported, then

$$J_{\text{real}}^r = -\beta H_{\text{real}}^r - (1 - \beta)(\ell_A + \ell_B).$$

Calling the positive entropy sum itself “ J ” is acceptable only if the table explicitly declares that J denotes the entropy component rather than the signed hybrid reward.

14 System-level illustrations

The following diagrams use the structure of the handwritten sketch: two robot trajectories, a shared-history boundary, an execution interval, the sensing updates, and the inference output. Solid arrows denote motion evolution; dashed colored links denote sensing information. The diagrams deliberately separate information used to create a robot’s posterior from hidden simulator ground truth used only to generate measurements.

14.1 Historical Visual-SLAM implementation

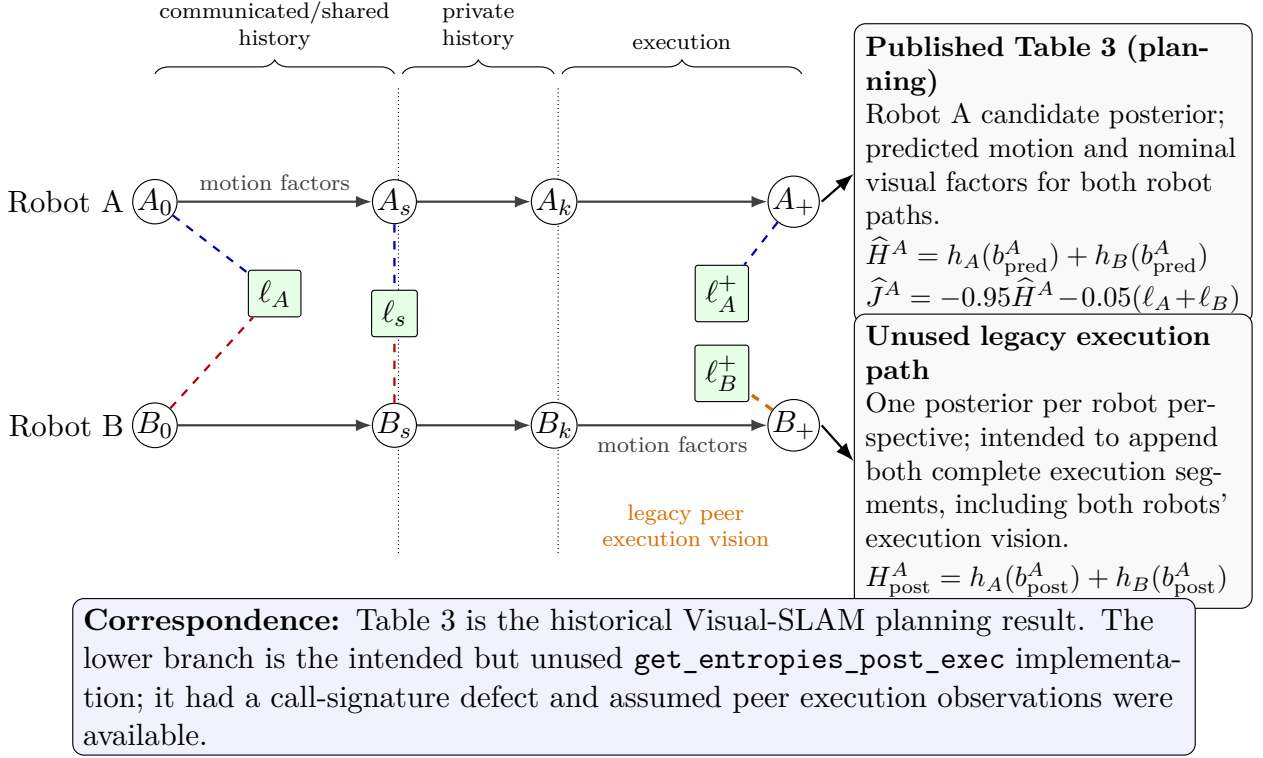


Figure 1: Historical Visual-SLAM planning and intended legacy execution logic. Blue and red dashed links are Robot-A and Robot-B vision factors; orange marks the peer-execution vision admitted by the legacy post-execution construction.

14.2 Moshe/Tanmoy search-and-rescue simulations

The search-and-rescue experiment is structurally simpler than Visual SLAM. The hidden state is a binary target-occupancy grid. Robot locations are treated as exactly known, so there is no pose posterior and no distinction between robot-specific pose entropy terms. Each agent evaluates the entropy of its own current belief over the *entire grid*:

$$H(b^r) = - \sum_i \sum_{x_i \in \{0,1\}} b^r(x_i) \log b^r(x_i).$$

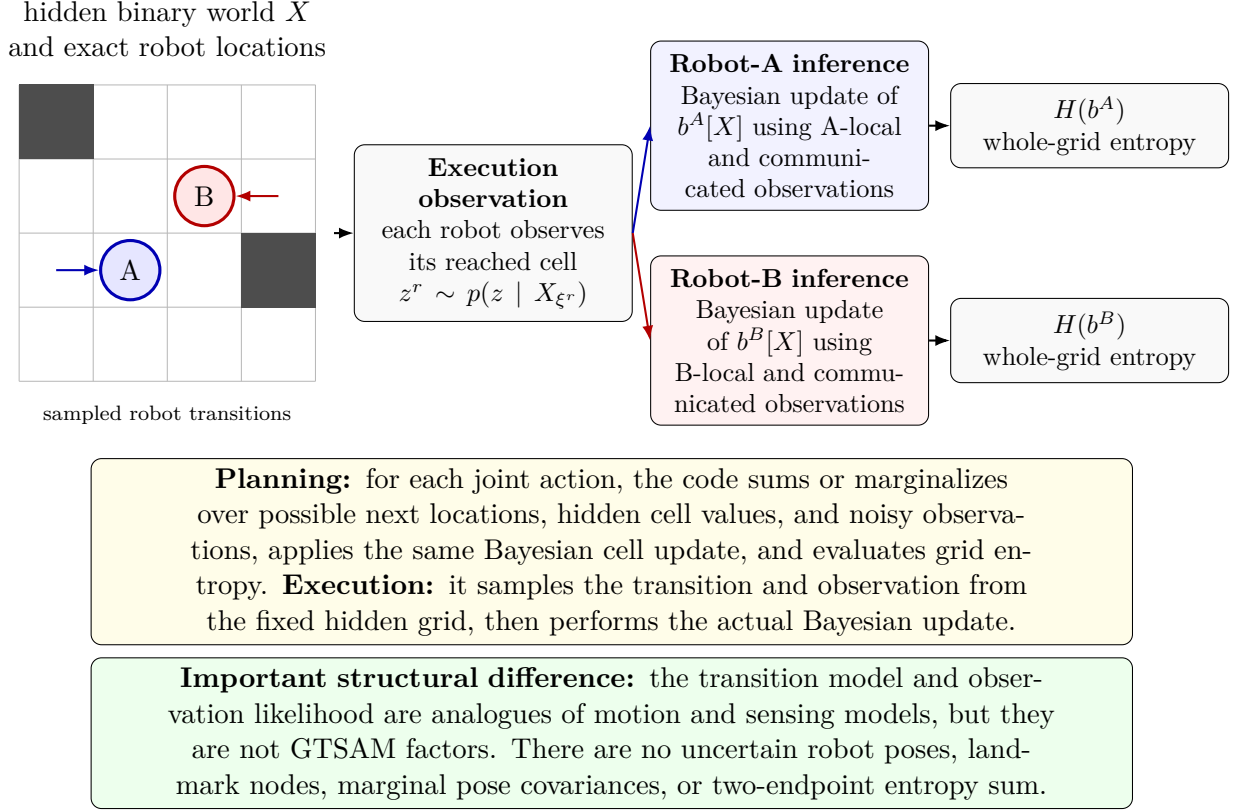


Figure 2: Moshe/Tanmoy search-and-rescue simulation, verified against the Julia simulation and the paper’s simulation section. Both robots reason about the same type of team state—the entire target grid—from different local beliefs.

14.3 Definitive current Visual-SLAM convention

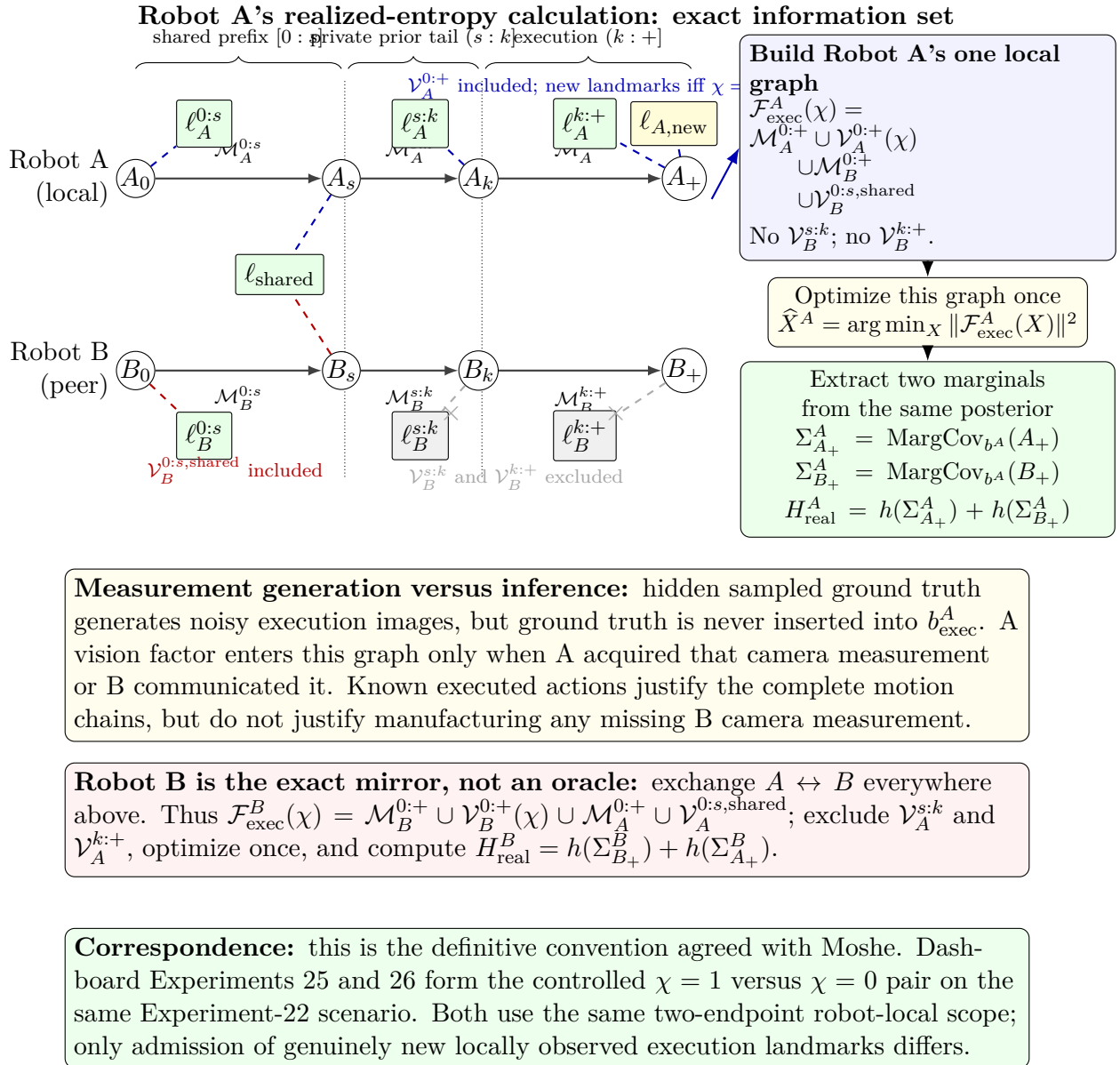


Figure 3: Exact Robot-A information set for the definitive current Visual-SLAM execution convention. Every black motion segment belongs to A's graph. Blue vision factors are A's complete local measurements, red factors are B's communicated prefix, and gray crossed factors are unavailable peer vision and must remain absent. Robot B uses the precisely mirrored construction.

15 Resolution status of Issues 11–15

For the experiment tables that compare Step-1 prediction with realized execution performance:

1. Report “Robot A realized team focused entropy” as H_{real}^A , containing both endpoint marginals from A’s execution belief.
2. Report “Robot B realized team focused entropy” as H_{real}^B , containing both endpoint marginals from B’s execution belief.
3. Optionally retain the decentralized self-entropy sum as a separately named system-performance statistic, but do not present it as either robot’s realized objective.
4. Do not call any of these sums “joint entropy.”
5. Define the Visual-SLAM specialization in the new manuscript as the sum of the final-position marginal entropy scores of all robots.
6. Validate factor provenance for every execution graph: actual local observations, actually shared peer observations, and known actions only. No ground-truth-generated factor may be inserted merely because it is available to the offline evaluator.
7. Report the newly discovered landmark switch χ for every result; use $\chi = 1$ as the primary real-execution metric and $\chi = 0$ as the known-landmarks-only ablation.

Issue 11’s candidate definition has been implemented in fixed copies of Experiments 22–24: each robot-perspective realized score contains both final pose marginals from one valid robot-local execution posterior. Issue 12’s historical implementation audit is also resolved by the evidence above: Table 3 used the same two-endpoint marginal-sum scope inside Robot A’s planning belief, while the separate legacy post-execution function intended the same scope inside each robot-local post-execution belief. Issue 13’s Moshe/Tanmoy simulation audit is now resolved: that experiment uses whole-grid binary entropy in each agent’s belief and has no pose-factor-graph analogue of the Visual-SLAM endpoint metric. Issue 15 is resolved by the definitive robot-local convention in Eqs. (6)–(9), agreed with Moshe on 9 September 2026. Issue 14—the numerical comparison of the original and corrected experiments, including the χ ablation—is the remaining calculation task.